Response to Reviewer

Annals of Data Science — Major Revision

Anomaly Detection Algorithm for Enhanced Fraud Identification in the Insurance Sector

Dear Editor and Reviewer,

Thank you for the constructive comments. We have applied the revisions within the original manuscript. Its five main sections, the preprocessing algorithm, the simple SAD algorithm, the multilayered SAD algorithm and the original detector/model families are retained. The changes clarify the contribution, reorganize the literature review and extend the experimental evaluation. Necessary implementation corrections are disclosed below. The package includes a clean manuscript, a version marking added and substantially rewritten passages in blue, a detailed change map, executable analysis code, and prediction-level outputs. The marked copy is a reading aid; the change map also identifies removed claims and maps the original sections and Algorithms 1–3 to their revised counterparts.

Reviewer comment. The manuscript must more sharply delineate the core innovations of SAD. The novelty of its "sequential" and "multilayer" structures is insufficiently highlighted; a comparative table versus state-of-the-art counterparts is recommended.

Response. Sections 1 and 3 now distinguish the mechanisms precisely. Sequential means that verified records from one batch augment training before predicting the next batch. Multilayer means that entropy is a gate applied after anomaly consensus, with entropy recomputed on the candidate subset. Neither mechanism is presented as a temporal sequence model or a reinforcement-learning policy.

The contribution statement identifies the review/update workflow and its empirical tests. The literature comparison table distinguishes supervised, unsupervised, active/reinforcement and hybrid/graph approaches by learning signal, strength and boundary. A second comparison table distinguishes SAD's mechanisms from recent optimized boosting, unsupervised deep learning, costly-verification RL and claims-graph methods. The two original workflow images are retained unchanged, and Algorithms 1–3 remain in the methodology section. An adjacent note clarifies how the executed protocol instantiates the conceptual diagrams. The submitted article's page layout, title block, section organization and ggplot2 visual conventions are preserved. Section 3 supplies the entropy definition, thresholds, training rules, feature handling and workload accounting. We present the contribution as a testable integration of existing components, not invention of the individual detectors or proof of universal superiority.

Reviewer comment. The literature review needs restructuring. The list-like survey should be reorganized into supervised, unsupervised, reinforcement-learning-based, and hybrid architectures, with the strengths, limitations, and research gaps of each class summarized explicitly.

Response. Section 2 has been rewritten into the requested four categories, followed by a synthesis of the research gap and comparator scope. Each category states its learning signal, practical benefit and limitation. We distinguish active acquisition from reinforcement learning and explain that SAD uses fixed review rules without a learned reward policy.

The bibliography now contains 40 cited references, including the method foundations needed by the restored detector and learner descriptions. The discussion incorporates recent insurance-specific work on systematic reviews, imbalance, optimized boosting and claims graphs. Three relevant *Annals of Data Science* sources support the discussion of imbalanced feature selection

(Kamalov et al., 2023), clustering (Xu and Tian, 2015), and financial-domain deep learning (Hassani et al., 2020). Their roles are explicit: the banking review is contextual, not an insurance-fraud numerical benchmark, and our Gini screen is not the supervised method of Kamalov et al.

Reviewer comment. The experimental evaluation is incomplete. Recent fraud-detection-specific baselines are missing, and no ablation study or validation on an independent second dataset is provided; adding these experiments would strengthen generalizability.

Response. Section 4 now reports new executed experiments on two benchmarks with ten fixed random seeds (1234–1243).

Additional empirical comparators. In addition to the eight component classifiers, we run balanced random forest and an adapted PSO-XGBoost comparator motivated by Ding et al. (2025). All classifiers use the same Simple-B acquired training records. PSO searches only within verified initial-partition records, with fold-specific SMOTE, and its chosen hyperparameters remain fixed in later batches. Section 4.4 gives the search budget and parameter bounds. The comparator is labeled "adapted" because we do not reproduce all data and implementation choices of the publication. This broadens the evidence while retaining a clear boundary: it is not an exhaustive reproduction of recent deep or graph-based systems.

Ablation and sensitivity. The Simple-A, Simple-B and Multilayer comparison studies entropy placement. A new Multilayer-static arm uses the identical initial reviewed records and fitted models but omits all later retraining. Its predictions use the same later claims, allowing paired seed-wise comparisons with updating Multilayer. A second complete Multilayer run uses $(T_{AD}, T_{ML}) = (5, 6)$ instead of $(3, 4)$. These two joint settings are described as sensitivity analysis, not an exhaustive grid or an isolated estimate of each threshold's effect.

Second benchmark. The supplied thesis archive contains Insurance Claims (1,000 records, 38 predictors, 247 recorded frauds), in addition to Carclaims (15,420 records, 32 predictors, 923 recorded frauds). The researcher identifies the source of the second file as the *Insurance Claims* dataset hosted by mexwell on Kaggle; this source is now cited in Section 4.1 and the bibliography. The analyzed CSV remains the supplied archival version. The same preprocessing rules, architecture definitions and SAD hyperparameters are used on both; preprocessing and models are fitted separately within each benchmark. The differently named `fraud_oracle.csv` is a duplicate representation of Carclaims and is not counted as an additional dataset.

On Insurance Claims, Simple-B has mean fraud F1 0.560 and AP 0.491; Multilayer has mean F1 0.497 and total review fraction 0.365. All revised scores, confusion counts and plots are regenerated from the new runs, rather than copied from legacy exports. Mean and sample SD across seeds are reported; we do not label them as independent-sample confidence intervals.

The second file has a different schema and record population, supporting workflow replication across benchmarks. We cannot establish the original insurer or real-versus-synthetic provenance of every field from the archive. We therefore avoid presenting this as prospective validation on a verified independent insurer portfolio. Section 5 states this remaining limitation, along with retrospective splitting and previously examined benchmark data.

Corrections arising from the reproducibility audit. The new implementation excludes the outcome by name before every detector, aligns row indices across methods, fixes candidate-to-original-row mapping and entropy gain, and forms acquisition masks before revealing outcomes. Training uses only verified records, including randomly audited non-alerts. SMOTE is implemented within training data, in accordance with the original method description. The revised pseudocode distinguishes accumulated real verified observations from newly oversampled training data and places retraining after the complete batch. Fraud is explicitly the positive class.

We distinguish alert rate from total review rate and report precision, recall, F1, specificity, balanced accuracy, ROC-AUC and AP. Stored warnings and computation times are retained without selectively discarding unsuccessful-looking split results.

The executed revised model implementation is Python. The empirical figures are rendered from those saved results using an executed R/ggplot2 script, with the same panel, colour, line and point conventions for both datasets. The original R scripts and outputs are archived for traceability. Each empirical figure has plot data, generation code, vector exports and a 720-dpi PNG. The two original workflow PNGs and their editable draw.io sources are retained unchanged. A numerical audit checks source-label alignment, split and review accounting, and independently recomputes the reported metrics from distributed prediction files. Figure labels and axis ranges have been adjusted for readability without changing numerical values. Section 4.6 now provides the public GitHub repository containing the data, executable analysis and plotting code, prediction-level outputs, figure assets and reproduction instructions: `https://github.com/Mehrab-Atighi/Anomaly-Detection-Algorithm-for-Enhanced-Fraud-Identification-in-the-Insurance-Sector`. Section 3.4 briefly acknowledges ChatGPT assistance with manuscript editing and revision of the analysis and plotting scripts; the authors retain responsibility for the code, results and final text.

Following the editorial instruction, the submission source archive contains editable LaTeX, bibliography files and PNG figure assets, with no PDF files. Compiled reading copies and the full computational archive are provided separately within the complete research package.

We hope the revised presentation and reproducible evidence address the reviewer's concerns while accurately describing the remaining limits.

Sincerely,
Mehrab Atighi, Sayed Jamal Mirkamali, and Amir T. Payandeh Najafabadi